

Air and Oxygen

Important Key Terms and Definitions

1. Components & Properties of Air

Active Component (Oxygen): The fraction of air responsible for carrying out vital natural processes such as respiration and combustion.

Inactive Component (Nitrogen): The major component of air that is chemically less reactive under normal conditions and slows down the activity of oxygen in the atmosphere.

Fractional Distillation: A process used to separate liquid air into its individual gaseous components based on their differing boiling points.

2. Atmospheric Structure & Pressure

Atmosphere: The layer of air surrounding the Earth up to an altitude of approximately 300 km.

Troposphere: The innermost atmospheric layer in contact with Earth's surface (0–10 km altitude), containing the majority of the atmosphere's mass.

Stratosphere: The atmospheric layer above the troposphere (10–50 km altitude) containing the protective ozone layer.

Mesosphere: The atmospheric layer (50–80 km) recognised as the coldest layer of Earth's atmosphere.

Thermosphere (Ionosphere): The high-temperature atmospheric layer where intense solar radiation ionizes gas molecules.

Exosphere (Magnetosphere): The outermost region of the atmosphere extending up to 10,000 km above the Earth.

Atmospheric Pressure: The downward pressure exerted by the weight of air per unit area. Its SI unit is the Pascal (Pa).

Barometer: An instrument used to measure atmospheric pressure, such as a Mercury Barometer or Aneroid Barometer.

Torricellian Vacuum: The almost complete vacuum space created above the mercury column inside a mercury barometer tube.

3. Chemical Reactions & Oxygen Concepts

Catalyst: A substance (such as manganese dioxide, MnO_2) that alters the rate of a chemical reaction without undergoing any permanent chemical change itself.

Combustion: An exothermic reaction accompanied by the evolution of heat and light, supported by oxygen.

Ignition Temperature: The minimum temperature to which a substance must be heated before it begins to burn in the presence of oxygen.

Inflammable Substance: A combustible material with an ignition temperature close to room temperature (e.g., petrol, kerosene).

Respiration: A slow biological combustion process in which oxygen combines with digested food (glucose) to release carbon dioxide, water, and energy.

Oxidation: A chemical reaction involving the addition of oxygen to an element or compound.

Oxyhydrogen Flame: A hot blue flame produced by ignited hydrogen and oxygen gas mixtures, utilized in high-temperature welding.

Acidulated Water: Water mixed with a small amount of sulphuric acid to allow electrical conductivity during electrolysis.

4. Corrosion & Prevention Methods

Rusting: The chemical corrosion of iron into hydrated ferric oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) due to prolonged exposure to moist air.

Galvanization: The process of coating iron sheets with a protective layer of molten zinc to prevent rusting.

Alloying: Mixing iron with other metals (like chromium and nickel) to create a homogeneous, rust-resistant mixture such as stainless steel.

Electroplating: Depositing a protective layer of another metal (such as nickel or chromium) onto iron using electric current.

Enamelling: Coating metal surfaces with enamel at high temperatures to withstand heat and moisture.

5. Special Mixtures & Uses

Carbogen: A mixture containing 95% oxygen and 5% carbon dioxide used for aiding breathing in medical emergency patients.

6. Air Pollution & Environmental Terms

Air Pollution: The alteration in normal atmospheric composition caused by the release of harmful or toxic gases.

Acid Rain: Rainfall made acidic by atmospheric oxides of nitrogen and sulphur (SO_2 , NO_2), which form nitric and sulphuric acids.

Greenhouse Effect & Global Warming: The trapping of heat radiated from Earth's surface by greenhouse gases (CO_2 , CH_4 , water vapour), resulting in a global temperature rise.

Ozone Depletion: The breakdown of the stratospheric ozone layer caused by pollutants like chlorofluorocarbons (CFCs).

Fly Ash: Fine unburnt solid residue produced during coal combustion in power plants, capable of causing health hazards.

Scrubber: An industrial device that uses lime water spray to precipitate and remove acidic oxides from flue gases.